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New Classification of Naturally Fractured Reservoirs Using Ronald Nelson's Method and Well Test Analysis

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Abstract

Classical well test analysis is limited in describing naturally fractured reservoirs when compared with the modern reservoir characterization methods, geological modeling and fault identification from new seismic acquisitions and modern logging tools. The early well testing fracture modeling techniques provided a basic fractures identification and description because it has not been updated with modern geo-mechanical understanding based on image logging tools and the detailed numerical grid properties based on new well testing software.

Later, Ronald Nelson's developed a geological study that combined the fluid dynamics with the static description of naturally fractured reservoirs based on the combinations of rock quality and fracture conductivity. This method classified naturally fractured reservoirs into four types, and it was helpful in understanding the fluid flow mechanisms, but it still needed to be more descriptive of natural fractures abundance, intensity, and connectivity in order to obtain stronger correlation between geology and reservoir engineering and well test analysis.

This paper presents a new classification of naturally fractured reservoirs as a proposal to upgrade Ronald Nelson's method with regard to pressure transient behavior studied from the well test analysis. Six different pressure derivatives shapes proves the presence of fractures and describes their different impact on well production, production sustainability and the general impact on fluid flow between reservoir segments. The well test analysis clearly indicates whether fractures are intersecting with the well bore or they are present at a distance from the well bore and also whether fractures are open conductive or sealing faults.

A new graphical cross plot is also introduced to show the relationship between well productivity as a function of rock matrix porosity and fracture transmissibility with six levels of matrix porosity/permeability and three levels of natural fractures conductivity and storativity. These parameters are plotted versus well productivity to give a clear visual method to understand reservoir performance and production sustainability.

The new description of reservoir performance under various types of fractures will help geologists and reservoir engineers to understand the structural aspects and fluid flow performance of naturally fractured reservoirs and assist in taking more effective decisions to improve field developments and reservoir management for maximum recovery.

Fractured Reservoir Description Using Modern Static and Dynamic Data

The recent advancement in seismic acquisition and data processing provided a good static reservoir description in identifying the presence of faults/fractures and mapping their lateral and vertical extent, their abundance, and connectivity (Singh et al., 2008). Fault interpretation from modern seismic attributes requires the integration with the Geo-mechanical modeling and regional stresses mapping from image logs and core studies in order to understand impact of fractures on the hydraulic communication between reservoir compartments and the general fluid dynamics of reservoir fluids prior to placing wells for field development (Geraral et al.,2003). Once wells are drilled and completed for production; the well test analysis is performed to study the pressure diffusivity into the reservoir by acquiring pressure data during well flow (drawdown) and well shut-in (pressure build-up) during drill stem tests (DST) or even during the reservoir production stage (Vasilev et al., 2016). The well test analysis of naturally fractured reservoirs has many values when integrated with geology, geophysics and geo-mechanics. The first value is to identify faults/ fractures presence. The second value is to know whether the faults are sealing or conductive and third value is to know whether they are intersecting with the wellbore, or they are at a distance. All these information can be induced from the pressure derivative shapes that can be calculated and plotted by the use commercial software (Buhidma et al., 2013; Spivey and Lee, 1999). After visually inspecting the derivatives shapes, we try to match the pressure derivative shapes with mathematical models in order to estimate matrix and fractures properties such as matrix and fracture conductivities (Deng et al., 2017), basic fracture geometry (fracture half length) and also to understand natural fractures abundance, and connectivity.

Fig. 1 shows different data sets that can be integrated to describe and characterize naturally fractured reservoirs such as geophysics, geomechanics and well test analysis.

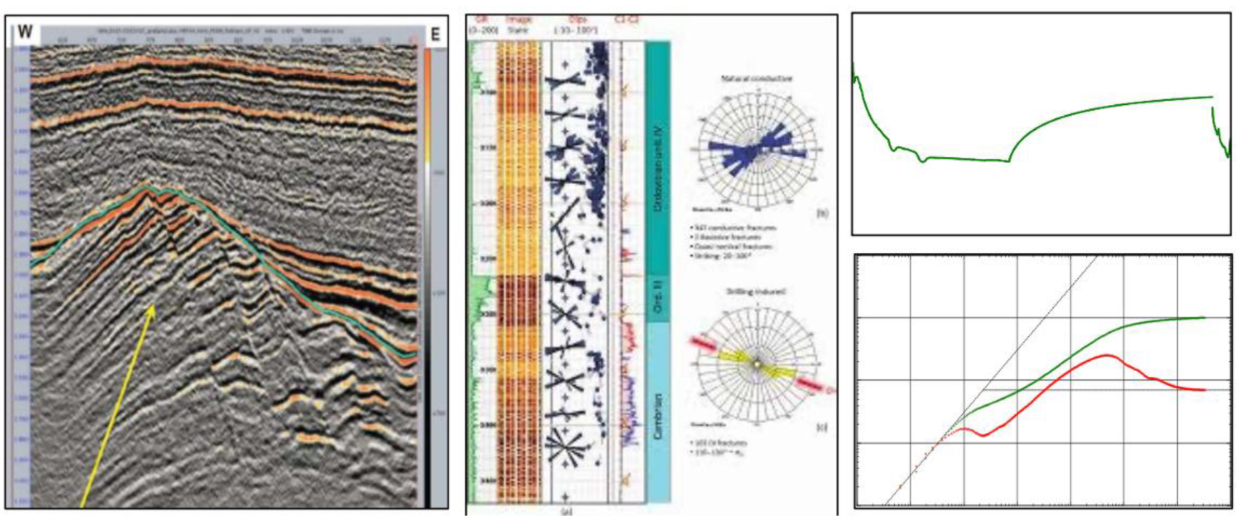


Figure 1—Modern seismic, geo-mechanics, and well test analysis

Ronald Nelson's Classification of Naturally Fractured Reservoirs

Ronald Nelson in (2001) described four different cases of reservoir fluid flow in naturally fractured reservoirs to explain the interaction between the matrix quality and fracture conductivity on the well production. Fig. 2 illustrates the dynamic response of fractures on well productivity by Ronald Nelson (1985). A cross-plot described the four-types of reservoir fluid flow in fractured reservoirs in order to explain the interaction between the rock porosity and permeability (Φ_{rock}, K_{rock}) and the fracture porosity and permeability ($\Phi_{fracture,}$ $K_{fracture}$) to determine the reservoir fluid flow whether purely Matrix flow or purely fracture flow or a combination of both matrix and fractures as follow:

- Type-1 is a high conductivity fracture in low porosity/permeability rock. The well production is attributed mainly to fractures while matrix slowly replenish the fracture flow. (Fracture flow)
- Type-2 is a medium conductivity fracture in fair porosity/permeability rock. The well production is mainly from fractures while matrix constantly replenish the fracture flow. (Fracture flow supported by matrix)
- Type-3 is a low conductivity fracture in good porosity/permeability rock. The well production is mainly from matrix while fractures enhance the well production. (Matrix flow enhanced by fractures)
- Type-4 is a zero conductivity fracture (sealing fault) in excellent porosity/permeability rock. The well production is mainly from matrix while the sealing fault creates a barrier between reservoir compartments. (Matrix flow).

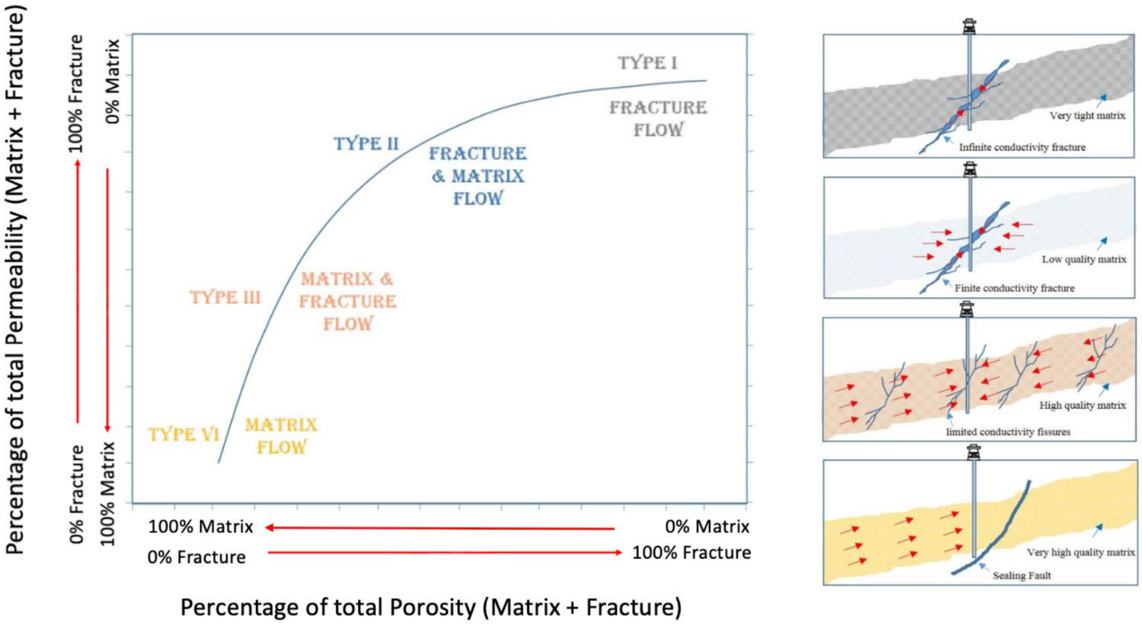


Figure 2—Ronald Nelson's classification of naturally fractured reservoir based on geological description

Fikri Kuchuk's Classification of Naturally Fractured Reservoirs Using Well Test Analysis

Kuchuck and Biryukov (2014, 2015) introduced a different approach to categorize fractured reservoirs with regard to reservoir simulation, production rate analysis and pressure transient analysis. Fractured reservoirs are described by their fractures conductivity, connectivity, storativity, and abundance. Fractures are either discrete (separated from each other) with low storativity or fractured are clustered in networks (connected together) with high storativity and conductivity. The four categories of fractured reservoirs are:

- Type-1: Continuously dual porosity fractured reservoir
- Type-2: Discretely fractured reservoirs
- Type-3: Compartmentalized faulted reservoirs
- Type-4: unconventional fractured basement reservoirs

Fig. 3 illustrates Kuchuk–Biryukov's (2014) classification of fractured reservoirs based on Pressure Transient Analysis (PTA) and flow regime interpretation. This approach categorizes naturally fractured reservoirs (NFRs) into 4 categories by examining the pressure derivative behavior in well tests and the orientation or distribution of fractures around the wellbore.

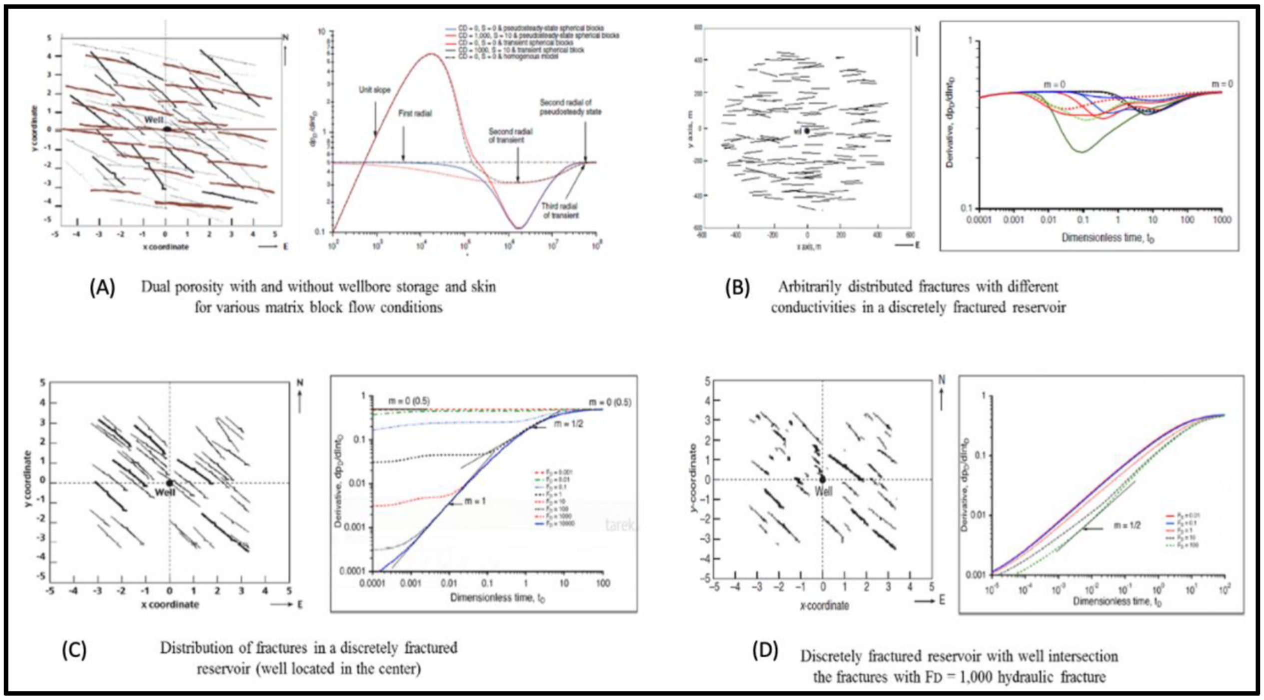


Figure 3—Kuchuk-Biryukov's classification of fractured reservoir based on pressure transient analysis patterns

The figure links fracture network geometry (isotropic/anisotropic/random/orthogonal) to Pressure Transient Analysis (PTA) flow regimes via dimensionless pressure through dimensionless pressure p_D and derivative $\frac{dp_D}{d\ln t_D}$. This allowing the identification of flow regimes and fracture network types. This classification provides a systematic framework for interpreting the re Type-1 (A): Continuously dual porosity fractured reservoir: fractures create a network, communicate hydraulically with each other and provide the overall conductivity (permeability) of the reservoir. The fluid volume contained in the rock (matrix storativity) is much higher than the fluid volume contained in the fractures (Fracture storativity). The storativity ratio can be estimated by using warren roots model (1963).

Type-2 (B): Discretely fractured reservoirs: fractures do not form a continuous conductive network. Only a limited of number of fractures communicate hydraulically with each other. Both fractures and rock matrix provide conductivity, but the overall storage capacity (porosity) is in the matrix.

Type-3 (C): Compartmentalized reservoirs: fractures are mineralized or clay smeared due to faulting will create a flow barrier and they divide the reservoirs into separated compartments.

Type-4 (D): is a zero conductivity fracture (sealing fault) in excellent porosity/permeability rock. The well production is mainly from matrix while the sealing fault creates a barrier between reservoir compartments. (Matrix flow).

The integration of both Ronald Nelson's method with Kuchuk-Biyoukov's method can be easily verified and developed further to show more types of fractures whether naturally or hydraulically developed. The study of many well test analysis cases of fractured carbonate and sandstone reservoirs proves that fractures can be divided into six types with distinctive shapes that provide a modern reservoir description that cab be proved by new structural geological models and new seismic attributes that can map fractures with great visual enhancement (Saputra et al., 2015). The need to generate a new classification of fractured reservoirs with a relationship between matrix quality and fractures properties (conductivity, geometry, abundance and connectivity) versus well productivity is found necessary to develop in order to honor both geological and well test analysis for a modern reservoir description.

New Classification of Naturally Fractured Reservoirs Using Ronald Nelson's Method and Well Test Analysis

This study presents a new classification of fractured reservoirs based on the study of numerous well test analyses of fractured reservoirs. The observation of these wells tests showed that fracture patterns on the log-log and derivative curve can be grouped into six patterns. Three types show the response of discrete disconnected fractures, and three types show the response of connected fracture networks. The six types of fractured reservoirs show different magnitude of well productivity and production sustainability mainly due to the following three elements:

Rock matrix quality has the key role in long-term well productivity and production sustainability. Matrix quality can be divided into a range of six levels to cover all possible rock types starting from extremely tight unconventional reservoirs up to the prolific Darcy reservoirs.

Fractures connectivity and geometry has an important impact on initial production rate and the magnitude of well productivity. Fracture networks (connected together) normally outperform discrete isolated fractures (separated from each other) for well productivity and sustainability.

Fracture conductivity has a strong impact on well productivity and fluid migration between reservoirs compartments. Fracture conductivity range can be divided into four levels starting from fully sealing faults with zero conductivity up to fault zones with open fracture corridors with infinite conductivity that give pipe flow.

Fig. 4 shows the six type's classification of fractured reservoirs divided into two categories and each category has three types sorted by their level of expected productivity as follows:

Category-1 (Discrete fractures with high conductivity and low storativity):

Type-1: Depleted discrete fractures in very low rock quality (unconventional reservoirs)

Type-2: Damaged discrete fracture with matrix support (leak skin, condensate drop-out, crushed zone)

Type-3: Linear/elliptical discrete fracture with matrix support (sustainable production)

Category-2 (Fracture networks with high conductivity and storativity):

Type-4: Fracture networks in tight matrix (Fractured basement)

Type-5: Fracture networks in permeable matrix (Double porosity)

Type-6: Fractures in fault zones with matrix support (Fault Zones)

This classification considers sealing faults as a third category (not included in the six types) due to their no impact on productivity as sealing faults create flow barriers (flow barriers) opposite to conductive fractures who work as flow enhancers.

The integration of well test analysis with the geo-mechanical model and modern seismic attributes for fracture detection and mapping is essential for all modern field studies in order to generate a valid reservoir model. A Fault model with all possible faults mapped and categorized with regional maximum and minimum horizontal stresses (σ_{Hmax}, σ_{Hmin}) identified is a prerequisite for all modern field studies workflow.

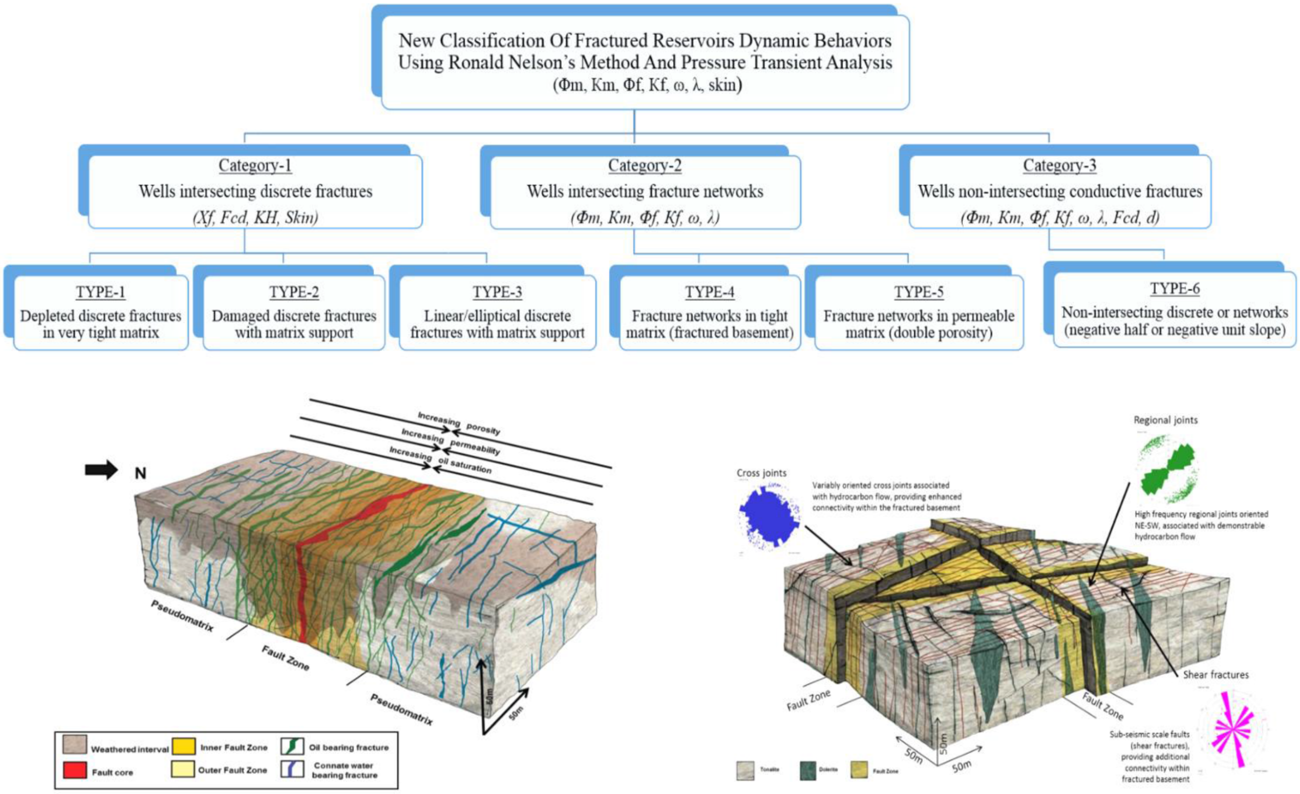


Figure 4—New classification of naturally fractured reservoir based on well productivity and well test analysis.

The following section presents a classification of pressure transient responses for different well–fracture interaction scenarios, modeled using numerical simulations. Fig. 5 through 10 illustrate six distinct dynamic reservoir types, ranging from single discrete fractures to complex fracture networks, each associated with unique pressure and derivative behaviors during well testing.

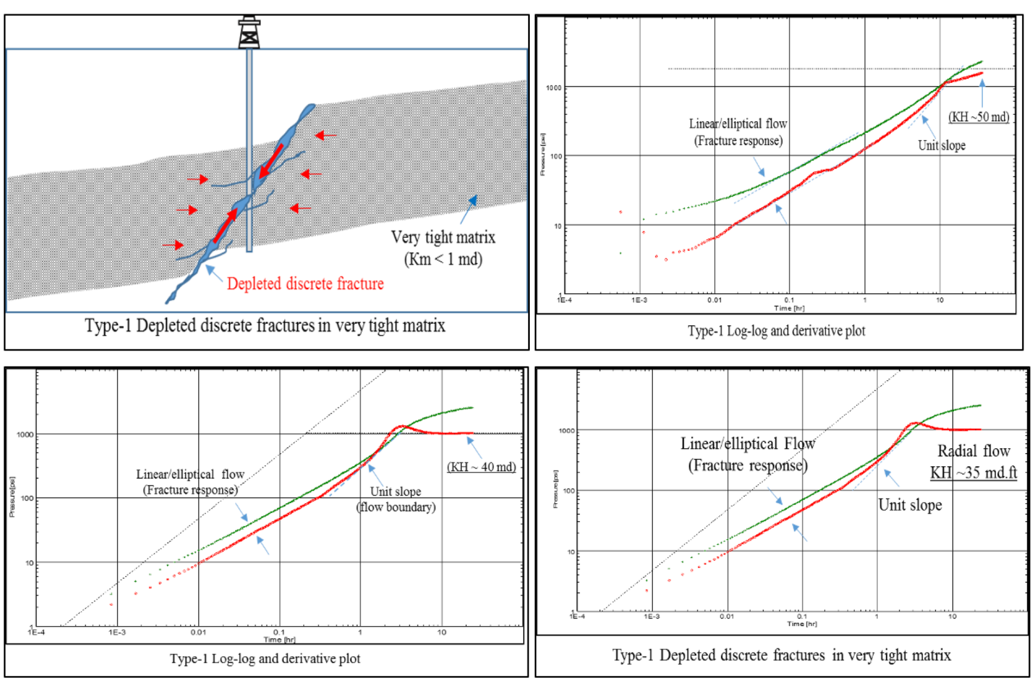


Figure 5—Dynamic response of Type 1: Well intersection depleted discrete fractures in tight matrix

Type-1: Well Intersecting a Depleted Discrete Fracture in Low-Permeability Rock

Fig. 5 illustrates the dynamic response of a well intersecting a single, depleted discrete fracture within a tight matrix rock. Such fractures may occur naturally or be induced through hydraulic fracturing. During well testing, a rapid pressure and production decline is observed, characteristic of depletion behavior, followed by slow pressure recovery during shut-in. The pressure derivative response shows a clear early-time fracture-dominated flow, followed by a positive unit-slope trend, indicative of fracture depletion and eventually transitions to late-time radial flow supported by the matrix. The reservoir's flow capacity (kh) is generally very low. The pressure data were modeled numerically as a proof of concept, with further development of analytical models recommended to capture this behavior accurately.

Type-2: Well Intersecting Damaged Discrete Fractures in Permeable Rock

Fig. 6 depicts a scenario where the well intersects discrete fractures that have been damaged, embedded in a moderately permeable matrix. These fractures, either natural or hydraulic, display low but sustained production rates. Fracture damage is categorized as:

Fracture skin: Localized near the wellbore, caused by Darcy or non-Darcy flow effects.

Leak skin: Develops along the fracture–matrix interface due to mechanical crushing or fluid leak-off.

During well flow, a significant drop in bottom-hole pressure is followed by a sharp build-up during shut-in. The derivative plot shows early-time fracture behavior, followed by a downward trend attributable to the impaired matrix permeability. Late-time radial flow is used to estimate matrix kh.

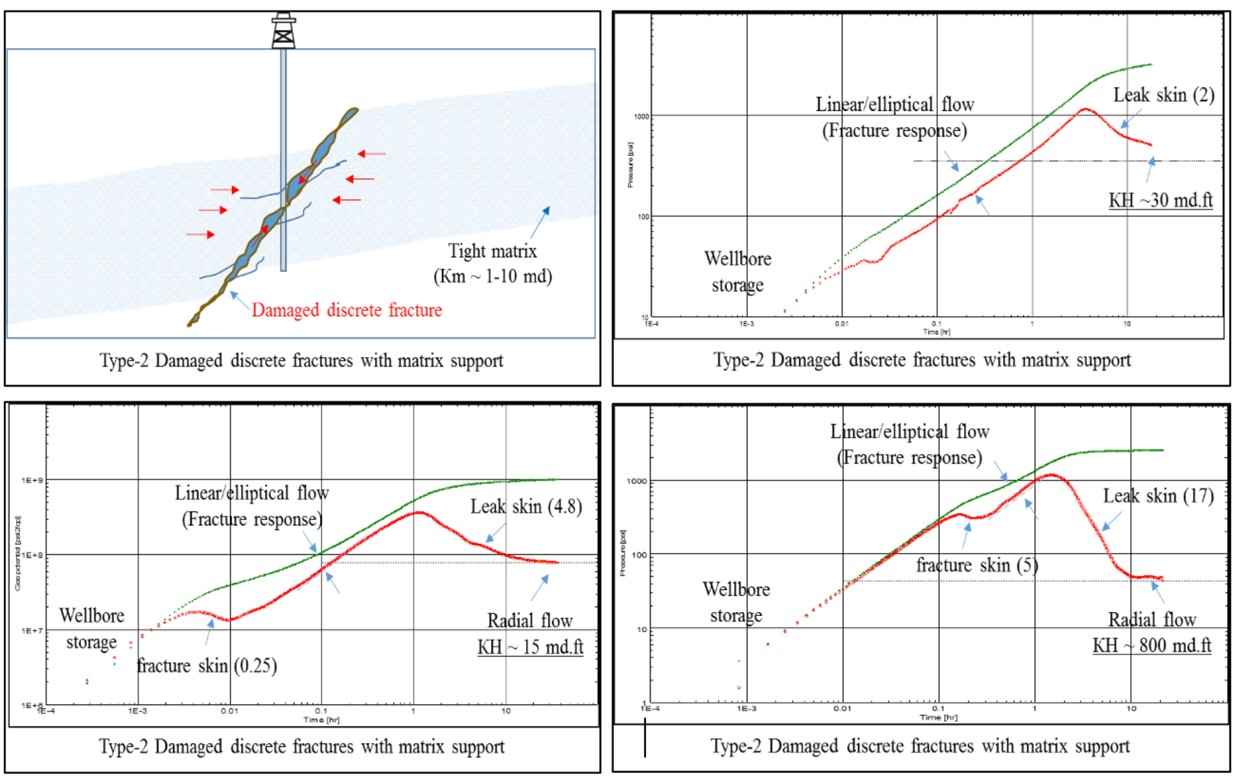


Figure 6—Dynamic response of Type 2: Well intersection damage discrete fractures in permeable rock

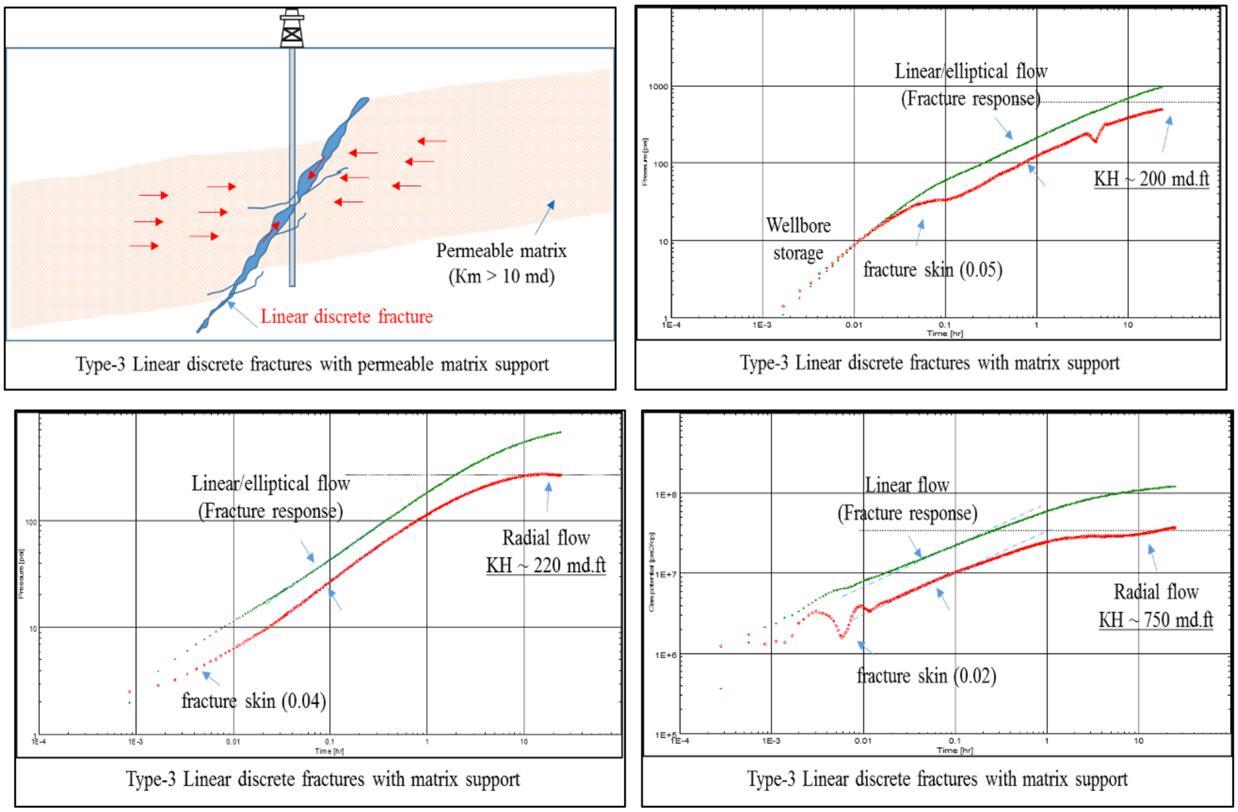
Type-3: Well Intersecting Discrete Fractures in High-Permeability Rock

Fig. 7 describes a well intersecting conductive discrete fractures in a high-permeability matrix. These conditions lead to high initial production rates with long-term sustainability. The production behavior is influenced by both matrix quality (porosity and permeability) and fracture geometry and conductivity (half-length X_f and fracture conductivity (FCD).

The derivative plot reveals

- Early-time linear or elliptical flow.
- Mid-time radial flow, reflecting strong matrix support.

Elliptical flow, with positive slopes ranging from 0.625 to 0.875, is especially evident when fracture conductivity or extent is limited.



Type-3 Linear discrete fractures with permeable matrix support

Type-3 Linear discrete fractures with matrix support

Type-3 Linear discrete fractures with matrix support

Type-3 Linear discrete fractures with matrix support

Figure 7—Dynamic response of Type 3: Linear discrete fractures with a matrix support

Type-4: Well Intersecting a Fracture Network in Low-Permeability Rock

Fig. 8 presents the response of a well intersecting a connected fracture network in very low-permeability rock, such as fractured basement formations. These reservoirs are often classified as unconventional and are only productive if well-connected fracture clusters are present.

Production rates are typically low but stable. The derivative curve shows a distinct early-time valley-shaped response, followed by horizontal radial flow at late time. The system can be analyzed using the Warren–Root dual-porosity model, enabling estimation of:

- The interporosity flow coefficient λ
- The storativity ratio ω
- The matrix flow capacity kh

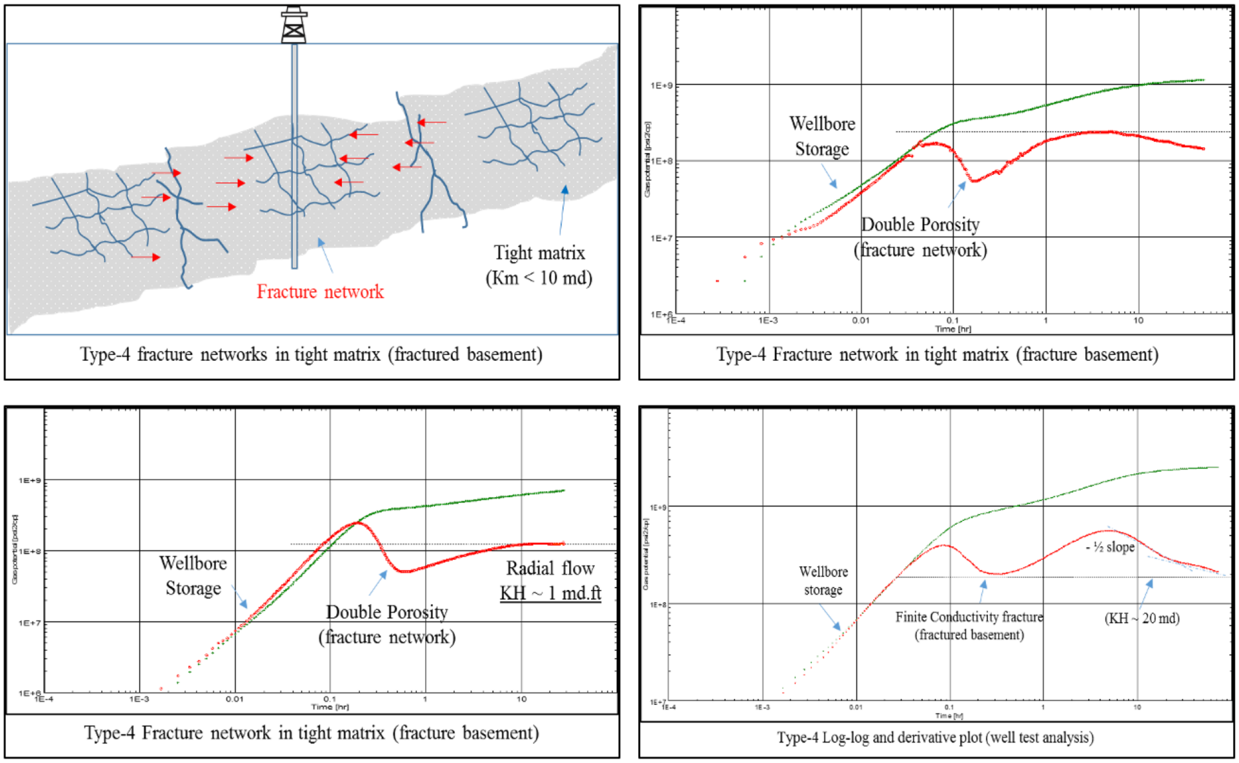


Figure 8—Dynamic response of Type 4: Fractures network in a tight matrix (Fractured basement)

Type-5: Well Intersecting a Fracture Network in Permeable Rock

Fig. 9 illustrates the Type-5 case, where the well intersects a connected fracture network within a permeable matrix, forming a true double-porosity system with contributions from both rock matrix and fractures. These wells demonstrate high, sustainable production rates with stable flowing pressure.

- The pressure derivative displays:
- An early-time valley, representing matrix-to-fracture interaction,
- A horizontal radial flow regime at late time.

The key distinction from Type-4 lies in fluid storage distribution:

- In Type-4, storage is dominated by fractures,
- In Type-5, both matrix and fractures store significant fluid volumes.

Accurate identification of this distinction is critical for optimizing drilling design and reservoir management strategies.

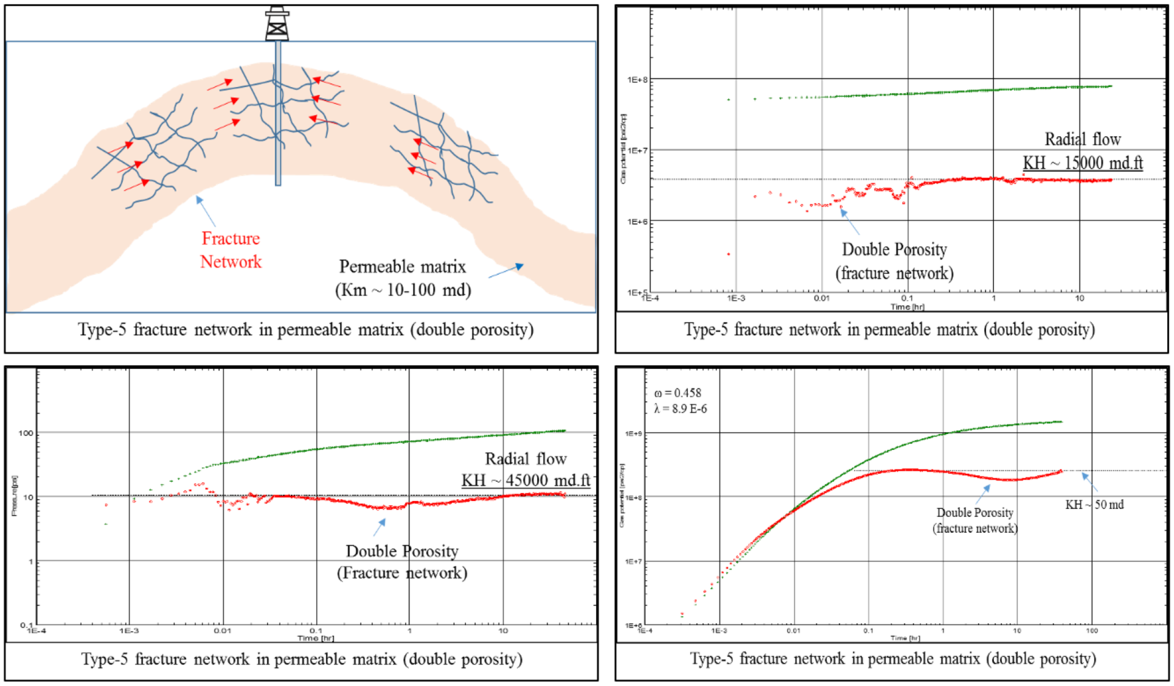


Figure 9—Dynamic response of Type 5: Fractures network in a permeable matrix (Double porosity)

Type-6: Well Near a Conductive Fault or Fracture Without Direct Intersection

Fig. 10 represents the Type-6 scenario, where the well is drilled adjacent to but not intersecting a conductive fault or regional fracture. These faults are identifiable from seismic surveys due to their large lateral and vertical extents and associated throws. Their hydraulic behavior—conductive or sealing is defined through geomechanical modeling and validated by dynamic testing.

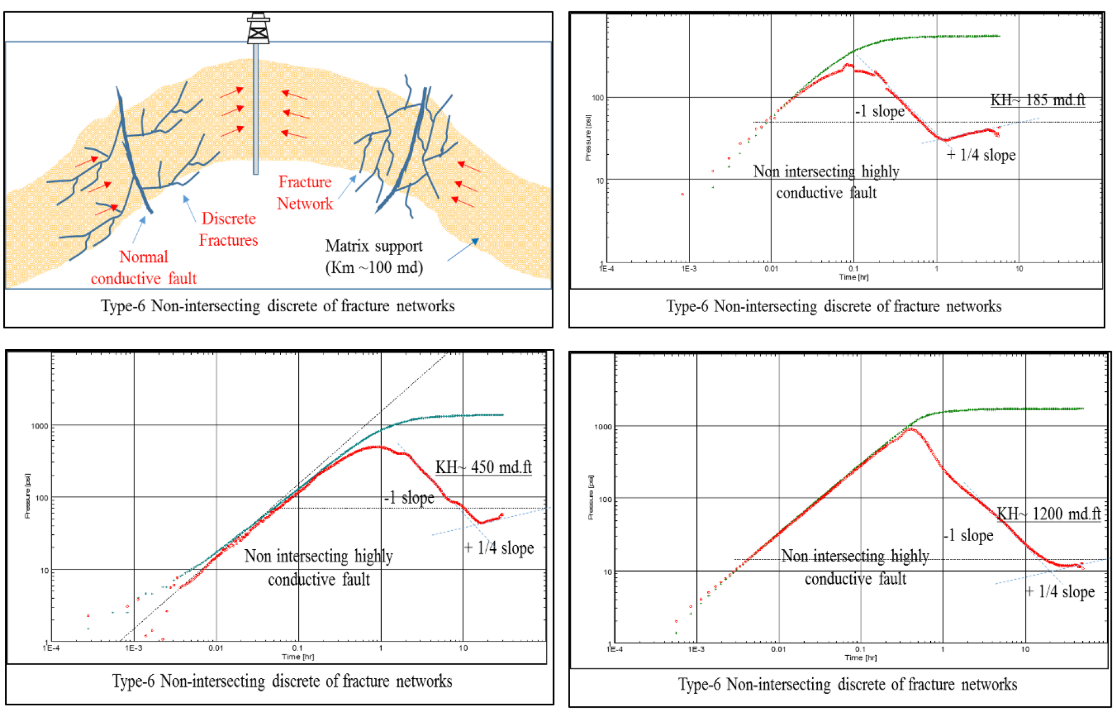


Figure 10—Dynamic response of Type 6: Non-intersecting discrete fractures or networks.

In high-permeability settings, such wells may achieve exceptionally high production rates with minimal drawdown, facilitated by micro-fractures or fissures that connect the well to the conductive fault. This results in a pipe-like flow regime, distinct from typical matrix or fracture flow.

The derivative curve shows:

- A sharp middle-time drop with a negative unit or half slope (−1 or −1/2),
- Followed by a positive quarter-slope (+1/4).

Conclusion

This study introduces a new classification system for naturally fractured reservoirs, integrating insights from classical geological frameworks (e.g., Ronald Nelson's porosity–permeability model) with modern dynamic well test analysis (e.g., Kuchuk–Biryukov's pressure transient interpretation). Traditional methods of fractured reservoir evaluation often lacked sufficient integration between geological structure, fracture connectivity, and dynamic flow behavior, limiting their effectiveness in guiding field development and reservoir management decisions.

By analyzing a broad range of well test datasets from fractured carbonate and sandstone reservoirs, six distinct pressure transient signatures were identified. These signatures represent unique combinations of:

- Matrix rock quality (porosity and permeability),
- Fracture geometry and connectivity (discrete vs. networked),
- Fracture conductivity and storativity, and
- Well-fracture interaction (intersecting or non-intersecting).

The classification was divided into two categories:
Category 1: Discrete fractures with high conductivity but low storativity, further divided into:

- Type-1: Depleted discrete fractures in tight matrix
- Type-2: Damaged discrete fractures with impaired matrix
- Type-3: Conductive fractures supported by permeable matrix

Category 2: Connected fracture networks with high conductivity and storativity, including:

- Type-4: Fracture networks in tight rock (e.g., fractured basement)
- Type-5: Double-porosity systems in permeable matrix
- Type-6: Highly conductive non-intersecting faults or fracture corridors

Each type shows a distinct derivative curve pattern, enabling diagnostic identification through pressure transient analysis. These patterns also correlate with expected production performance, offering a predictive framework for well deliverability and long-term reservoir behavior.

Furthermore, a graphical cross-plot model was developed to visualize reservoir productivity as a function of matrix porosity–permeability and fracture conductivity–storativity. This tool allows intuitive performance prediction and provides a robust platform for integrated reservoir characterization.

By synthesizing static data (from seismic, image logs, and core-based geomechanics) with dynamic well test results, this new classification offers a practical and comprehensive approach to:

- Distinguish between sealing and conductive fractures,
- Estimate the contribution of matrix vs. fractures to flow,
- Identify the nature of fracture–wellbore interaction, and
- Guide drilling, completion, and production optimization.

This integrated methodology enhances the understanding of fluid flow mechanisms in fractured reservoirs and supports data-driven, geomechanically-informed decision-making in field development planning. The framework is particularly valuable in complex and unconventional reservoir systems, where traditional single-domain models often fail to capture the true reservoir behavior.

References

1. Buhidma, I. M., Habbtar, A., & Rahim, Z. 2013. "Practical Considerations for Pressure Transient Analysis of Multi-Stage Fractured Horizontal Wells in Tight Sands". Paper SPE-168096 presented at *SPE ATS&E* held in Al-Khobar, Saudi Arabia, 19-22 May 2013.
2. J.P. Spivey, W.J. Lee 1999. "Variable wellbore Storage Models for a Dual-Volume Wellbore". Paper SPE-56615 presented at SPE ATCE held in Houston, Texas, USA, 3-6 October 1999.
3. Kuchuk, F., Biryukov, D., Fitzpatrick, T., and Morton, K. 2015. "Pressure Transient Behavior of Horizontal Wells Intersecting Multiple Hydraulic and Natural Fractures in Conventional and Unconventional Un-fractured and Naturally Fractured Reservoirs", paper SPE-175037 presented at *SPE ATCE* held in Houston, Texas, USA, 28-30 September 2015.
4. Kuchuk, F. and Biryukov, D. 2014. "Transient Pressure Test Interpretation From Continuously and Discretely Fractured Reservoirs". *SPE Res Eval and Eng* **17** (1): 82–97. SPE-158096 PA
5. Gerard B., El Deeb M., Hussein B., and Frederic C. 2003. "*Seismic facies analysis for fracture detection: A powerful techniques*". DOI:10.2118/81526-MS, June 2003.
6. Deng Q. Nie R., Jia Y. et al 2017. "Pressure transient behavior of a fractured well in multi-region composite reservoirs", *Journal of Petroleum Science and Engineering*, Vol.-**158** (2017) pages 535-553.
7. Ronald A. Nelson 2001. "*Geologic Analysis of Naturally Fractured Reservoirs*". 2nd edition published in 2001 by Butterworth-Heinemann.
8. Ronald A. Nelson 1985. "*Geologic Analysis of Naturally Fractured Reservoirs*". 1st edition published in 1985 by Butterworth-Heinemann.
9. Saputra, R., Hidayat, A., & Rahmanto, W. A. (Saka Indonesia Pangkah Limited) 2015. "Adapting Fracture Corridors and Diffuse Fractures in Single Porosity Model of Ujung Pangkah Carbonate Reservoir". Paper SPE-176101presented at SPE Asia/Pacific Oil & Gas Conference and Exhibition held in Bali, Indonesia, 20-22 October 2015.
10. Singh S. K., Abu-Habbiel H., Khan B., Akhbar M., Etchecopar A., and Montaron B., 2008. "*Mapping fracture corridors in naturally fractured reservoirs: An example from Middle East carbonates*". Published in first break Vol.-**26** (2008), pages 109-113. DOI: 10.3997/1365-2397.26.1119.27999
11. Vasilev, I., Alekshakhin, Y., and Kuropatkin, G. 2016. "Pressure Transient Behavior in Naturally Fractured Reservoirs: Flow Analysis". Paper SPE-182562 presented at SPE Annual Caspian Technical Conference & Exhibition held in Astana, Kazakhstan, 1-3 November 2016.